



SAURAV ACHARYA

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Summary

Research Associate at the Center for Environment and Energy Development (CEED India) with experience in integrated energy system modelling for Jharkhand, and biomass resource assessment. Previously completed M.Tech thesis research at IIT Bombay on large-scale power system modelling of India using PyPSA. Listed among the contributors powering PyPSA on its official platform. Research interests include energy system modelling, renewable energy integration, energy storage, and sustainable power system planning.

Experience

Research Associate – Energy

Center for Environment and Energy Development (CEED India)

July 2025 – Present

- Conducted integrated energy system modelling for the Jharkhand power sector using PyPSA. (Under review)
- Assessing sub-district level biomass resource potential across Jharkhand using spatial and agricultural datasets for bioenergy planning. (Ongoing)

Research Intern (M.Tech Project)

Indian Institute of Technology Bombay

June 2024 – June 2025

- Performed techno-economic modelling of India's power sector with focus on Battery Energy Storage System (BESS) and Pumped Hydro Energy Storage (PHES).
- Analyzed and optimized the duck curve for India through the integration of energy storage.

Education

M.Tech in Energy Technology

Tezpur University

Aug 2023 – Jun 2025

PG Diploma in Space and Atmospheric Science

Physical Research Laboratory (PRL), Ahmedabad

Oct 2021 – Jun 2022

M.Sc. in Physics

Dibrugarh University

Aug 2019 – Sep 2021

B.Sc. in Physics

North Lakhimpur College

Aug 2016 – Jun 2019

Research Projects

Working Paper: Solar Curtailment and Renewable Energy Curtailment Analysis in India

- Ongoing research assessing solar and wind curtailment across India's five RLDC regions using Python, with a focus on quantifying curtailed energy, estimating financial losses, and analyzing regional curtailment patterns.

Study of Dust Optical Depth over Twelve Martian Years

Physical Research Laboratory, Ahmedabad

2021 – 2022

- Analysed dust optical depth from MY24-MY35 using the LMD Mars Climate Database.
- Identified global dust storm events with a 4–5× increase.

Technical Skills

Energy Systems: Energy modelling, renewable energy transition, energy storage

Tools & Software: PyPSA, pvlib

Data Analysis: Python, MS Excel